

Name: Shreyans Tatiya

Roll no: 16010123325

Batch: C3/ DV Fri

TITLE: Visualize hierarchical and network data

Objective:

To visualize hierarchical and network data.

Expected outcome of Experiment:

CO2	Detect and understand the stories within datasets and its applications.
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Books/ Journals/ Websites referred:

< The students should mention clearly the books, websites or journal papers, and articles they have refereed to while completing the task>

Background Theory:

This experiment focuses on the visualization of hierarchical and network data to enable effective analysis of structured relationships and interconnections within datasets. Hierarchical data represents information organized in parent–child relationships, where entities are arranged across multiple levels, such as organizational structures, file systems, product categories, or geographical divisions. Network data, on the other hand, represents complex relationships between entities where elements (nodes) are connected through links (edges), commonly found in social networks, transportation systems, communication networks, and recommendation systems.

Visualizing hierarchical data using techniques such as tree diagrams, treemaps, and decomposition trees allows users to understand data distribution, dominance of categories, and drill-down relationships across multiple levels. These visualizations make it easier to identify patterns such as heavily populated branches, imbalanced hierarchies, or dominant subgroups that may not be apparent in tabular data.

Network data visualization employs node-link diagrams and force-directed graphs to illustrate relationships, interactions, and connectivity among entities. By encoding properties such as node size, color, and edge thickness, network visualizations help in identifying influential nodes, clusters, hubs, and isolated components. Such representations are particularly useful for understanding interaction strength, relationship density, and structural dependencies within the

dataset.

Visual analysis of hierarchical and network data supports data-driven decision-making and storytelling by transforming complex relational structures into intuitive graphical forms. Interactive visualization features such as drill-down, filtering, and highlighting further enhance exploratory analysis, enabling users to uncover hidden insights, compare relationships, and interpret large-scale structures efficiently. These techniques are widely applied in business intelligence, social media analysis, cybersecurity, supply chain management, and organizational analytics.

Task to be performed on the given dataset:

Visualize Network Data

I. Prepare Your Excel Data

Your Excel data needs a structure that defines relationships. You need at least two columns:

Source: The starting point of a connection (e.g., a customer, a webpage, an employee).

Target: The destination point of a connection (e.g., another customer, another webpage, a manager).

Weight (Optional): A measure to indicate the strength or thickness of the connection (e.g., number of sales, page views)

II. Import the Excel Data into Power BI

Open Power BI Desktop.

Go to the Home tab and select Get Data > Excel.

Browse to your Excel file, select it, and click Open.

In the Navigator pane, select the table(s) or worksheet(s) you want to use, then click Load.

III. Import the Force-Directed Graph Custom Visual

The **force-directed graph** is a custom visual and is not available by default.

In the Visualizations pane, click on the three dots (...) and select Get more visuals.

In the pop-up window, search for "force" or "network".

Locate the Force-Directed Graph visual (developed by Microsoft) and click Add.

The new visual icon will appear in your Visualizations pane.

IV. Create the Force-Directed Graph

Click the Force-Directed Graph icon in the Visualizations pane to add a blank visual to your report canvas.

Enlarge the visual to an appropriate size.

From the Data pane, drag your fields into the visual's wells in the Visualizations pane:

Drag your Source field into the Source data role.

Drag your Target field into the Target data role.

Drag your Weight field (e.g., sales, views) into the Weight data role. This will make the connecting lines thicker for stronger relationships.

V. Customize the Visual

In the Visualizations pane, select the Format your visual (paint roller) tab.

Explore formatting options such as:

Links: Enable arrows, adjust thickness, and change link color.

Nodes: Change node colors or adjust their size.

Data labels: Turn on/off or change the text size and color.

You can now interact with the graph by dragging the nodes around the canvas to explore the relationships between different entities

Visualize hierarchical data

Step 1: Load the Data

1. Open Power BI Desktop.
2. Click **Home > Get Data > Excel Workbook**.
3. Select your file and click **Load**.

Step 2: Select the Treemap Visual

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1. In the **Visualizations** pane on the right, look for the icon that looks like a cluster of rectangles.
2. Click it to create a blank Treemap on your canvas.

Step 3: Map the Fields

Drag fields from the **Data** pane into the buckets in the **Visualizations** pane as follows:

- **Category (Group):** Drag **Region** first, then drag **Category** right below it.
 - *Why:* This creates a "Drill-down" hierarchy. You will see Regions first, and you can click them to see Categories.
- **Values:** Drag **Sales (\$)**.
 - *Why:* This determines the **size** of the boxes. Larger sales = Larger box.
- **Details:** (Optional) Drag **Product** here if you want to see individual product boxes without drilling down (caution: this can make the chart very busy).
- **Tooltips:** Drag **Profit Margin (%)**.
 - *Why:* When you hover over a box, you will see the profit margin.

Step 4: Add Color Coding (Conditional Formatting)

To make the chart smarter (e.g., darker color = higher profit):

1. Go to the **Format your visual** tab (paintbrush icon).
2. Expand **Colors** > Click the **fx** button next to "Default color" (or "Colors").
3. **Format style:** Gradient.
4. **What field should we base this on?:** Select **Profit Margin (%)**.
5. **Minimum:** Red (Low profit).
6. **Maximum:** Green (High profit).

How to Interact with it

- **Drill Down:** Because you put both Region and Category in the "Category" well, you can click the "Drill Down" arrow (single down arrow) in the visual header, then click on "North America" to zoom in and see the specific categories inside North America.

Implementation screenshots:

Visualize Network Data

Step 1: Network Visual Creation

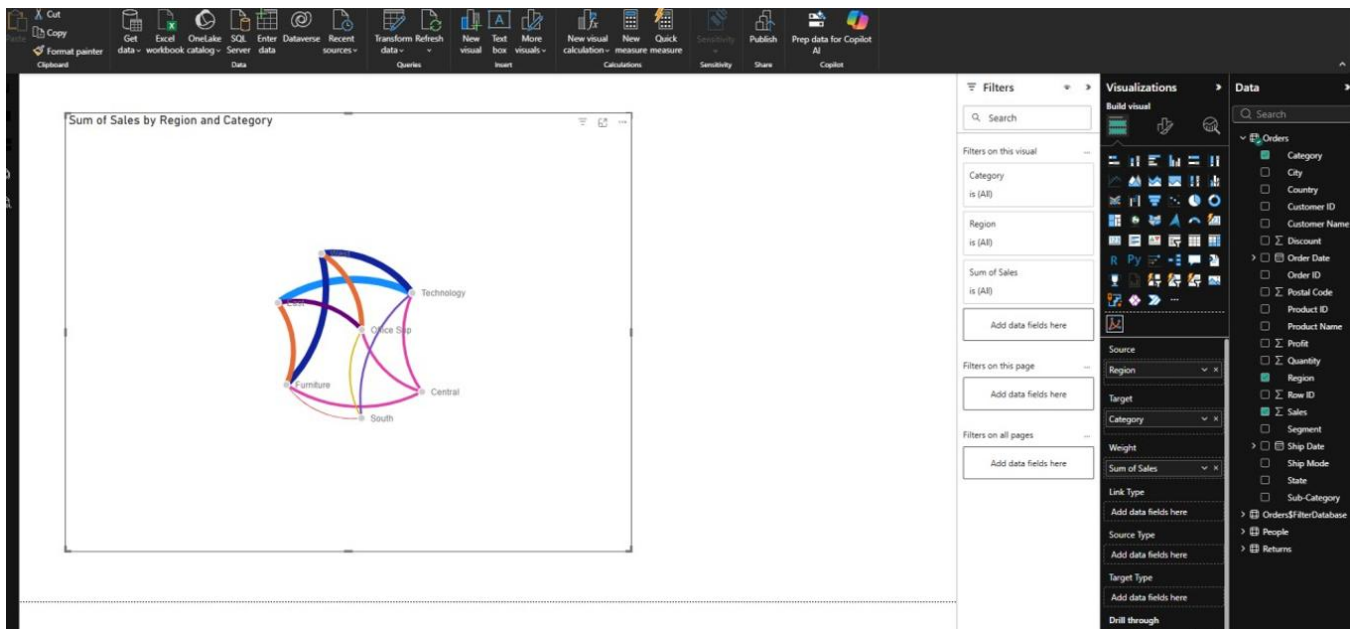
A network graph was created in Power BI Desktop to show relationships between **Region** and **Category**.

Step 2: Field Mapping

- Source → Region
 - Target → Category
 - Weight → Sales
- This defines the connections and their strength.

Step 3: Structure of Graph

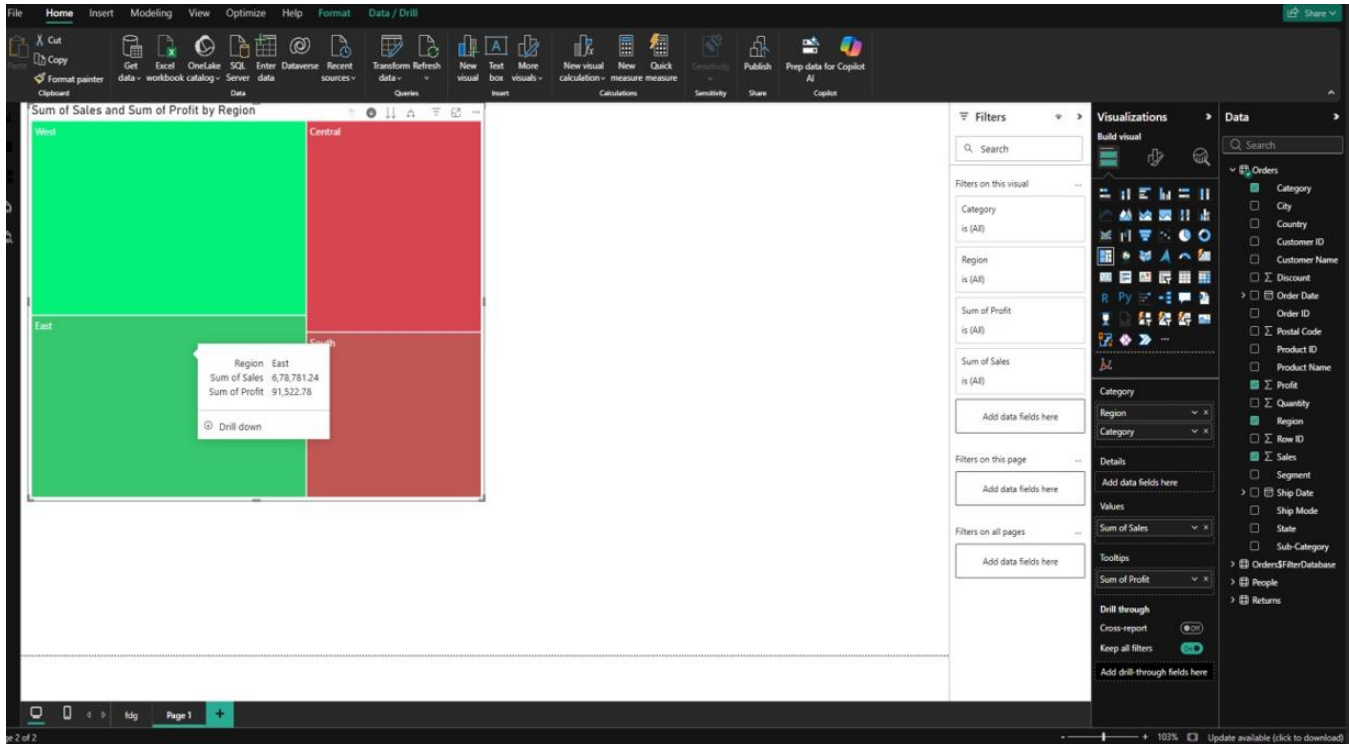
Nodes represent regions and categories, while links show their relationships. Thicker links indicate higher sales.



The graph highlights strong region-category combinations and helps identify high-performing areas.

Visualize hierarchical data

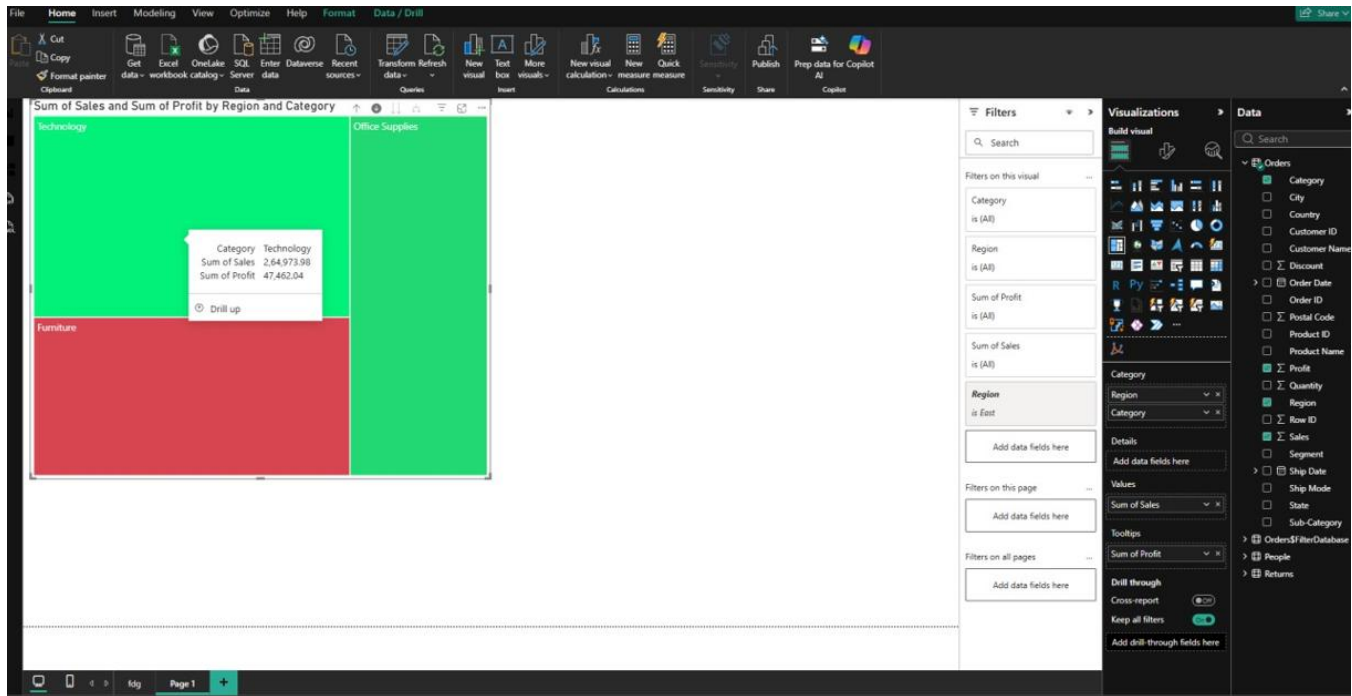
Step 1: Treemap by Region



Explanation:

A treemap was created in Power BI Desktop to display Sales by Region. The size of each box represents total sales, while color indicates profit (green = high, red = low).

Step 2: Drill Down to Category



Explanation:

Using the drill-down feature, the visualization shows **Category-wise distribution within East region**. This helps analyze which product categories contribute most to sales and profit in a specific region.

The treemap highlights:

- **Larger boxes = higher sales**
- **Green areas = profitable segments**
- **Red areas = low-profit segments**

This helps quickly identify strong and weak performing regions and categories

Conclusion:

This experiment demonstrated how to visualize both **network** and **hierarchical data** using Power BI Desktop. The network graph helped in understanding relationships between regions and categories, while the treemap effectively showed sales distribution and profitability. Overall, these visualizations made it easier to identify key patterns, high-performing areas, and business insights.

Post Laboratory Questions

1. Why are tree and treemap visualizations suitable for hierarchical datasets?

Tree and treemap visualizations are ideal for hierarchical data because they clearly show **parent–child relationships** in multiple levels (e.g., Region → Category → Sub-category). They help in understanding the structure of data, comparing groups, and identifying which segments contribute more using size and layout. Treemaps additionally use area to represent values, making them effective for showing proportions within the hierarchy.

2. What challenges arise when visualizing large networks?

Visualizing large networks can lead to **visual clutter**, where too many nodes and edges overlap, making the graph difficult to read. It becomes hard to identify meaningful patterns or relationships. Additionally, large networks may cause **performance issues** like slow rendering, and require filtering or clustering techniques to simplify the visualization.

3. In which scenarios would a network visualization be preferred over hierarchical visualization?

Network visualization is preferred when the data involves **complex, many-to-many relationships** instead of a fixed hierarchy. It is useful in scenarios like social networks, recommendation systems, supply chains, or customer-product relationships, where entities are interconnected and do not follow a strict top-down structure.